Hyeongjin Kim

1301 Paresky Center
Williamstown, MA 01267
+413 (664) 1697
hk6@williams.edu

Education

2017–2021 **Williams College**, *Williamstown, MA.*
Bachelor of Arts in Physics; Cumulative GPA: 3.97; Major GPA: 4.00
Dean's List: All Graded Semesters

2010–2016 **Kristin School**, *Auckland, New Zealand.*
International Baccalaureate Diploma; IB Score 44/45
Class of 2016 Valedictorian

Research Experience

Jan 2020 – Present **Strauch Lab: Optimization and Noise Analysis of Quantum Logic Gates**.
Williams College Physics Department
Honors Thesis Student, Research Assistant
Advisor: Frederick W. Strauch, Professor of Physics

- Investigated a system of two parametrically coupled, fixed-frequency (7 GHz and 5 GHz) transmons with *symmetric* anharmonicities (300 MHz) to perform the $\sqrt{\varphi\text{–SWAP}}$ gate.
- Analytically minimized the leakage of $\sqrt{\varphi\text{–SWAP}}$ gate by forming a discretized sequence of pulses, giving gate errors of $\sim 10^{-5}$ for **any** gate times as fast as 6.5 ns.
- Currently working on a paper to present our work on analytical/numerical pulse optimizations and circuit synthesis of various unitary gates for our transmon systems.

June 2019 – Aug 2019 **Strauch Lab: Optimization and Circuit Synthesis of Quantum Logic Gates**.
Williams College Physics Department
Research Assistant
Advisor: Frederick W. Strauch, Professor of Physics

- Investigated and numerically simulated/optimized to perform a fast and high-fidelity gate (termed $\sqrt{\varphi\text{–SWAP}}$) for a system of two parametrically coupled, fixed-frequency (7 GHz and 5 GHz) transmons with *asymmetric* anharmonicities (400 MHz and 200 MHz.)
- Developed a circuit that implements the CNOT gate with two $\sqrt{\varphi\text{–SWAP}}$ gates and seven single-qubit gates, allowing a fast implementation of any arbitrary two-qubit gates.
- Numerically optimized a two-quadrature pulse that achieves gate errors between 10^{-4} and 10^{-6} of $\sqrt{\varphi\text{–SWAP}}$ for gate times between 5 ns and 10 ns.

Feb 2018 – Dec 2018 **Jensen Lab: Dynamics of Adhesive Wetout**.
Williams College Physics Department
Research Assistant
Advisor: Katharine E. Jensen, Assistant Professor of Physics

- Investigated the mechanics of adhesive contacts of rigid glass spheres with silicone gel surfaces of varying Young's modulus via an co-independently developed two-dimensional image analysis MATLAB program.
- Synthesized silicone gels of varying stiffness, prepared probes with rigid glass spheres, and used an optical microscope with a high-speed camera to investigate adhesive dynamics.
- Assisted in the construction of a Michelson interferometric system for the three-dimensional imaging of high-speed interface dynamics.

Presentations

Aug 2019 Williams College Summer Science Tea Talk and Poster, "Fast and High-Fidelity Quantum Logic Gates for Parametrically Coupled Transmons."

Aug 2018 Williams College Summer Science Tea Talk and Poster, "Dynamics of adhesive wetout and detachment."

Jul 2018 UMass Amherst Soft Matter Day, "Dynamics of adhesive wetout and detachment."

Teaching Assistantships

Spring 2020 **Algorithm Design and Analysis**, *Williams College Computer Science Department*.
Hosted homework sessions and grade assignments on efficient, provably correct algorithms on divide-and-conquer, dynamic programming, NP-hardness, and randomized approximation methods.

Spring 2019 **Mathematical Methods for Scientists**, *Williams College Physics Department*.
Hosted homework sessions and graded assignments on ordinary and partial differential equations, special functions (including Bessel and Legendre), and numerical methods in Mathematica.

Community Engagement

Jun 2020 – Present **Co-Chair**, *Williamstown, MA*.
Williams College Society of Physics Students
- Organizing and hosting departmental events for physics students.
- Mentoring and helping underclassmen in physics.

2018–2019 **Finance Committe Member**, *Williamstown, MA*.
Williams College Council
- Analyzed budgets and constructed optimal funding strategies for College Council.

Honors and Fellowships

Aug 2020 Phi Beta Kappa Induction: Top 5% of the class by cumulative GPA.

2018 – 2020 Williams College Summer Science Research Fellowships: Funded summer research experience supervised by Katherine Jensen (2018) and Frederick Strauch (2019 & 2020).

Relevant Coursework

Physics Classical Mechanics; Controlling Quanta; Quantum Physics; Electromagnetic Theory; Vibrations, Waves and Optics; Electricity and Magnetism; Mathematical Methods for Scientists; Modern Physics.

Mathematics Measure Theory and Hilbert Spaces; Representation Theory; Real Analysis; Probability; Applied Partial Differential Equations; Abstract Algebra; Linear Algebra; Discrete Mathematics; Multivariable Calculus.

Computer Science Algorithm Design and Analysis; Data Structures and Advanced Programming; Introduction to Computer Science.

Other

Languages Python, Java, LaTeX

Tech Mathematica, MATLAB

Hobbies Badminton, Rubik's Cube, Video Games, Book Collection